

Completing the programme: verified results and release

M24 • 02 Nov 2028 - 27 Nov 2028 • 75 hours

The M24 objective

Bring the learning outputs together into an engineering project: verified requirements, reproducible data, versioned code, CAD and configurations, a technical report and a defence. The original plan's outcome is **open release v1.0, a technical report and project defence**. This month, prepare a release candidate whose results can all be checked. The owner decides on public publication separately after reviewing the contents and comments.

The principal devices are the M23 six-legged robot and the robotic hand from the separate workstream after M17. Each retains its own requirements, limits, calibrations and evidence. A shared portfolio index links their software components but does not substitute one project for the other. If either device has not passed mandatory acceptance, state its exact status and completion blockers; presentation does not turn an unfinished device into a finished one.

What to gather before the first session

Prepare M1-M23 outputs: source code, CAD, drawings, circuits, calibrations, configurations, simulators, tests, datasets, failure logs and alternative solutions. Verify that files exist, rather than relying on links in an old report. For the six-legged robot, obtain standing, wave and ripple, slope, failure-response and energy data. For the hand, obtain evidence for the grasps, force and current limits, sensors and responses actually included in its OPT project.

Do not begin the month by rewriting the entire architecture or changing documentation tools. Select a pinned working version, describe its boundaries and fix problems that prevent verification of claimed functions. Your software-development experience helps with automation, but the number of unit tests cannot replace physical validity.

What constitutes a professional result

Every important claim traces to a requirement, procedure, configuration, raw measurement and reproducible calculation. State separately what simulation, rig tests, the physical machine and external review establish. Include unsuccessful approaches and limitations in the report: when causes and conclusions are justified, they demonstrate engineering judgement.

M24 ends on 27.11.2028; the original calendar then schedules G3 for 27.11-10.12.2028 and G4 for 10-22.12.2028, 35 hours each. These hours and the hand's separate 120 OPT hours are outside M24's 75 hours. By 22 December 2028, both systems must have an explicit final status. Study hours and a portfolio alone do not confer a degree, senior-engineer qualification or research mastery.

75 hours: verification campaign, documentation and defence

Time allocation

Hours 1-20: 10 hours on requirements, metrics and the verification plan; 10 hours auditing data and conducting the first reproducible runs. **21-40:** 10 hours on safe states, failures and release contents; 10 hours on repeated checks, versioning and clean reproduction. **41-60:** 10 hours on technical explanation and external-review preparation; 10 hours on the report, diagrams, defence and review comments. **61-75:** 5 hours choosing a professional focus and final criteria; 10 hours closing comments, checking the portfolio and preparing the G4 handoff.

Total: 35 hours of theory and analysis and 40 of practice. The first three theory blocks each contain five 2-hour sessions; the last is 2 + 2 + 1. Each 10-study-hour practice block includes setup, testing, processing, corrections and evidence recording. The 30-45-minute external meeting is included in the third practical, not added to the budget.

Organising the package

Create one index covering requirements, configurations, tests, raw data, calculations, report, instructions, comments, release and portfolio. Give requirements stable IDs, such as HEX-STANCE and HAND-GRASP, and configurations unique revisions. Do not link evidence only to a filename such as final.csv: include run_id, file hash and system version. Large data need suitable storage with integrity checks; DVC is useful if it already fits the project.

The first-page dates define the original plan's window. Shared boundary dates, holidays and separate G blocks do not create extra hours. Place numbered work within available sessions without double-counting. Use time awaiting a reviewer's response for independent tasks; external review is incomplete until a response arrives.

Priorities when time is short

First resolve invalid measurements and untested mandatory functions, then an unreproducible build, then ambiguous instructions and reviewer comments. Website design, complex animation and an additional demonstration video come last. A narrower verified feature set may be released with explicit limitations; lowering a mandatory criterion after failure requires declaring the requirement change.

Keep two lists: v1.0 release blockers and development after the programme. For each new feature, determine whether it is needed to verify behaviour already claimed. If not, add it to the future specialisation. This preserves time for both physical projects and final integration.

Requirements, metrics and links to measurements

What to study

Distinguish verification, checking fulfilment of a stated requirement, from validation, checking that the system solves the intended task under real conditions of use. A specific trial checks 120 seconds of standing; assessing whether the robot makes repeatable experiments convenient also requires instructions, preparation and user evaluation. Select a method for each requirement: analysis, inspection, demonstration or testing, sometimes in combination.

A good requirement identifies the object, action, conditions and criterion. Replace low latency with the events between which the interval is measured, the clocks used, the load and the limit. Derive the limit from project dynamics, frequency and loading; a round number alone is not justification. A specification citation identifies the page, units and applicable conditions.

Task 1. A requirements matrix for both systems

Inputs are the six-legged-robot objectives, hand objectives and actual release candidates. 1) Break broad claims into verifiable requirements with stable IDs. 2) For each, specify origin, method, criterion, admissible configuration and required log. 3) Assign PASS/FAIL/NOT TESTED/NOT APPLICABLE with an explanation. 4) Verify that every feature on the portfolio homepage appears in the matrix.

Save requirements_traceability.csv. Check that every mandatory requirement has a specific procedure and result source; NOT TESTED is not a pass. Boundary case: after actuator replacement, an old calibration and test remain linked to the new assembly. Update links and repeat affected checks. There is no need to retest an unchanged M1 parser merely to increase the test count.

Task 2. Audit metrics and units

Calculate latency as $t_{\text{end}} - t_{\text{start}}$ only after establishing clock relationships. Define jitter explicitly, for example as deviation of cycle period from its target; show its distribution, p50, p95, p99 and observed maximum with sample count. For angular error, state conventions and units. Obtain energy by integrating $U \cdot I$ over time; use E/d in J/m only for a valid non-zero distance. For the hand, define grasp, holding and release success separately.

Save metrics_spec.md and a calculation script applied to one reference dataset with a known answer. Check that values and units agree across code, table and plot; do not present small-sample quantiles as precise estimates of rare extremes. Boundary case: a current-recording interval is missing. Mark energy incomplete or use an explicitly justified estimate with an interval; do not fill the gap with zero.

Independent stopping, failures and recovery

What to verify

Verify the final physical state, not only a software enumeration changing to STOPPED. Removing torque may make a walking robot fall; for a hand, it may release or trap the object depending on mechanics. Each system needs a described path for normal stopping, emergency termination of force or torque application, and recovery. Support and motion restraint come from the design verified on earlier rigs.

Independence applies to specific failures. If a hardware path depends on the same board's power or firmware, show that in the circuit. A second thread in the same process does not make stopping independent of process failure. Test in the intended protected configuration with an observable physical indication of the result and a criterion established in advance.

Task 3. Reconfirm the final state

Use the final six-legged-robot and hand revisions. 1) Trace command and power paths on the circuit. 2) Prepare robot support and a safe test object or fixture for the hand. 3) Repeat normal stopping and the independent emergency path with the programme operating normally. 4) Repeat the permitted rig case with the control process stopped. 5) Check button release and restored communication: the previous command must not resume automatically.

Save `safe_state_evidence` for both devices with a timeline and observation of the mechanism. Acceptance: each claimed path actually reaches the described state within justified time and displacement limits. Boundary case: the log reports deactivation but the actuator still applies force or torque. The test is FAIL; investigate the hardware path and repeat. Software confirmation does not replace measurement.

Task 4. Failure, diagnostics and recovery

From the risk register, choose sensor loss, stale telemetry, a stopped node and a configuration error. Check them first in replay or HIL. For each, record stimulus time, detection, restricted behaviour and operator action. Physically reproduce only permissible variants using the established procedure; destructive battery failure and short circuits are unnecessary for verifying software responses.

Save `fault_recovery_matrix` and a minimal regression set. Check that after a node recovers, the system rechecks calibration, mode and data freshness before authorising motion. Boundary case: different errors need different responses, but one reset button hides both. Separate confirmation that the cause is resolved from permission for new motion. Retain failed cases and the rationale for protective behaviour.

A reproducible code, CAD and data package

What reproducible means

Another engineer must receive consistent source code, hardware documentation, configuration and sufficient data to reproduce the main results. Code, CAD, circuit and calibration have separate versions linked by a shared release manifest. An STL without its source parametric model and dimensions does not explain the design; a binary without its configuration version does not define controller operation.

Distinguish reproducing calculations from saved data from conducting a new physical experiment. The former can be checked in a clean environment without the robot. The latter requires an assembly, equipment, calibration and admissible conditions. Instructions must state which results readers can obtain without hardware and which are confirmed only on your unit.

Task 5. The v1.0 candidate manifest

Inputs are the pinned revisions of both systems. 1) List source code, CAD, circuits, BOM, calibrations, dependencies, tests, data, documentation and demonstrations. 2) State each item's version or hash and purpose. 3) Separate raw data from derived plots. 4) Check for dependencies on local paths and ensure required external files are accessible through the instructions.

Save `release_manifest.json` and a local candidate archive. Check that every link resolves, checksums match and release notes describe this exact version. Boundary case: calibration comes from a different hand despite matching file format. Include the physical unit's serial identifier and revision and add a compatibility check. A filename does not establish correct configuration.

Task 6. Reproduce in a clean environment

Create a separate working copy or isolated environment, restore pinned dependencies and only the dataset included in the candidate. Follow the instructions from the start: build, software tests, selected-log processing, table and plot generation. Documentation scripts must not automatically connect powered hardware. Hardware instructions require a deliberate mode choice and entry checks.

Save `reproduction_log.md` with commands, environment, runtime and omissions found in the instructions. Check that at least one key plot for each of the six-legged robot and hand reproduces the same substantive result within the stated tolerance. Boundary case: a notebook displays an old saved-cell plot while calculation fails. Clear saved outputs, run from scratch and fix the dependency. A missing raw-data file blocks the claim that relies on it.

For M20-M21 results, include the QP configuration, solver-failure check, unchanged classical baseline, random-number generator seeds and the OOD/fallback report. Identify learned components investigated only in simulation. Increased reward does not verify physical operation; explicitly record lack of hardware authorisation in the report and portfolio.

Technical report and external project defence

What you must be able to explain

Organise the report around the problem each system solves: objective, requirements, architecture, methods, experiments, results, limitations and future work. In the architecture, show physical components, command flow, measurement flow and power circuit. A separate timing diagram explains the measurement-to-action path; a state diagram explains motion authorisation, stopping, failure and recovery. Do not combine everything into one unreadable diagram.

In methods, describe procedure, instruments, calibrations, initial conditions and processing. In results, separate measured facts from hypotheses. Each plot has units, a caption, version and attempt count; photographs show the fixture. Limitations must name surfaces, loads, durations, uncertainties and untested modes rather than offer a generic paragraph about future potential.

Task 7. A report that stands on its own

1) Write a half-page account of each system's result. 2) Draw three diagrams: architecture; timing; states and stopping. 3) Select 4-6 main plots that substantiate requirements. 4) Trace each back to the raw file. 5) Analyse one unsuccessful design choice and one trade-off, such as speed versus heating or grip force versus protecting the object.

Save technical_report and diagram source files. Check that a reader can identify the operating envelope and locate data supporting the main claim within a few minutes. Boundary case: body-tilt and surface-slope plot axes share the same label. Correct notation and caption, then recheck the text. Every plotted quantity must have an unambiguous meaning.

Task 8. A 30-45-minute external review

Prepare a compact package in advance: objectives, diagrams, requirements matrix, one successful trial and one failed trial. Choose someone able to assess at least part of the mechanics, control or software system. During the meeting, allow 5 minutes for context, 10 for architecture and limitations, 10 for evidence and 5-15 for questions. Record each objection, affected requirement, required action and closure method.

Save design_review_minutes with the comments actually received. Check that at least one technical objection is addressed substantively and the response is supported by data or an explicit limitation. Polite praise does not replace engineering review. If the person has not replied, status is pending; you may rehearse independently, but cannot call it an external defence. The owner deliberately sends invitations and grants access; no automatic mailing is required.

A shared test campaign for two devices

Specification of the integrated work

Create campaign.yaml with conditions and run_id values. For the six-legged robot, test standing, wave and ripple, levelling, restricted operation without a camera and stopping. For the hand, test the project's intended grasps, holding, release, detection of an unreliable sensor and failure response. If the hand is not mounted on the body, do not claim integrated mobile manipulation: these are two completed devices sharing a verification method.

Freeze version, success definition and trial unit first. For the walking robot, the unit is a complete bounded scenario with restored initial conditions; for the hand, a specified cycle on a known object with defined position and load. Reprocessing one file checks calculation reproducibility; it does not add physical trials. A different hardware revision starts a separate result group.

The 30-50+ series and limits of inference

The overall plan suggests 30-50+ tests or a justified sample size. Plan the sample around the claim: comparable attempts to estimate success rate and separate boundary conditions for coverage. M23 data may count if configuration, procedure, conditions and criteria match the final ones; this reuses evidence rather than adding newly conducted experiments. Define the hand's series separately, without pooling grasp successes with walking.

An example arrangement is 40 attempts for one frozen six-legged-robot task and a separately agreed grasp series. If time is insufficient, narrow the claim and justify N, or move the open block to G4 within the overall schedule. Do not promise a reliability percentage from an arbitrarily small sample. Every mandatory mode retains its test status even if excluded from the main homogeneous series.

Execution and calculation

First check instruments on a known signal and run a pilot trial. Then follow a fixed order or alternate modes to control for discharge and heating. After every attempt, record outcome, initial conditions and interventions. Classify failures as mechanics, power, communication timing and bus, perception, control, procedure or data. Categories support repair but do not remove failures from the final table.

For success rate, report k/N, the confidence-interval method and applicability conditions. If heating creates serial dependence, do not call individual cycles independent; analyse blocks or restrict conclusions to descriptive statistics. For continuous metrics, show the distribution across attempts, units and actual observation duration. Observed maximum current and the hardware's admissible maximum are different numbers from different sources.

Tests of the evidence package itself

Six required scenarios

T1 - a different configuration. In a test copy, substitute calibration from another unit: hardware activation must be refused or the report must flag incompatibility. **T2 - corrupted raw data.** Change one byte or row in a copy: integrity checks or the parser must detect the problem and must not present the result as unchanged.

T3 - a missing time interval. Delete part of a log copy: duration, energy or latency calculations must indicate incompleteness. **T4 - an old notebook plot.** Generate the report after clearing saved cell outputs: the plot must be regenerated from data rather than retained from a previous version. **T5 - control-process failure.** On an admissible rig, repeat independent stopping and prevention of automatic resumption.

T6 - reproduction without undocumented local dependencies. In a clean copy, run the claimed recorded-data analysis without hardware: instructions must work with the specified files rather than an undocumented author directory. Explicitly identify any access requirement and provide an accessible example dataset. Save expected outcome, actual result and package version for each test.

Checking whether claims are justified

Take five main claims from the README and CV. Trace each chain: claim → requirement → test → configuration → raw data → calculation. If a link is missing, restore evidence, narrow the claim or remove it. A matching trial supports 120-second standing in this configuration; it does not support stable operation in every condition. A claim to have developed a controller and compared it with a baseline must identify your own contribution and comparison conditions.

Distinguish certification, educational verification and operational authorisation. Your tests establish specific project requirements; mentioning safety does not make them official certification. Name completed checks directly: independent disconnection, command limiting and response to data loss. Do not claim more than the methods support.

Closing defects

For every defect, record reproduction, consequences, correction and affected requirements. After changing the system, repeat the smallest set that demonstrates the correction and absence of related regressions. A caption edit does not require rerunning the entire two-year plan; a power-circuit or calibration change requires the corresponding physical check.

The change log identifies results belonging to rc1, rc2 and the final version. Label earlier-version data retained as comparisons. Release with a known limitation is acceptable only when it does not conceal an unmet mandatory requirement for a claimed function and is clearly documented in the instructions.

Hours 1-40: requirements, testing and reproduction

Hours 1-10: analytical preparation

1-2: inventory results, check both devices' status and choose test candidates. 3-4: requirements, verification versus validation and the start of task 1. 5-6: definitions of latency, jitter, energy, errors and grasp success from task 2. 7-8: sampling plan, failure classification and compatibility of earlier data. 9-10: trial order, stopping criteria, instruments and run-description fields.

By the end of hour ten, prepare campaign.yaml and the requirements-status table. Assess remaining work by open rows rather than plot count. If the physical hand is missing, this is a separate blocker to overall programme completion: M24's 75 hours are for verification and presentation, not an unannounced replacement of its 120-hour construction work.

Hours 11-20: the first practical

11-13: task 1, traceability checks and raw-file audit. 14-15: task 2, reference inputs and reproduction of main metrics. 16-18: pilot trials and planned campaign repeats on verified configurations. 19-20: failure classification, observation register and a decision on remaining data needs. Do not run a new physical mode merely because old data are incomplete; prepare its procedure first.

Some earlier-module results may be unusable because timestamps are absent or configuration changed. Do not retrospectively alter raw data. Retain the original, state the reason and collect new measurements or narrow the claim. Synthetic test data are useful for checking calculations but do not replace physical evidence.

Hours 21-40: stopping and the package

21-22: paths to final states for both systems and independence from specific failures. 23-24: recovery, stale commands and reactivation. 25-26: release contents and the links between code, CAD, firmware and calibrations. 27-28: data versions, checksums and a clean environment. 29-30: release notes and reproduction instructions without undocumented assumptions about the author's environment.

31-33: task 3, verify stopping of both systems in suitable fixtures. 34-35: task 4, failure and recovery in HIL or on the rig. 36-37: task 5, manifest and local rc1 archive. 38-40: task 6, clean reproduction and instruction fixes. The deliverable is a package a reviewer can use without spoken guidance at every step. Grant others access deliberately after selecting the package contents.

Hours 41-75: defence, corrections and release readiness

Hours 41-50: explaining the engineering result

41-42: report structure and each system's main claim. 43-44: architecture and paths for energy, commands and measurements. 45-46: timing diagram and failure/recovery states. 47-48: presenting results, uncertainties and unsuccessful approaches. 49-50: questions for the external reviewer and defence rehearsal. Each session must improve a specific part of task 7, not merely increase text length.

Arrange the external review with the chosen person in advance. Prepare a small package and a precise question in their area of competence. If the meeting moves, continue the report and independent tests; external-review status remains open. Correspondence and invitations are deliberate actions by the project owner; the plan does not generate them automatically.

Hours 51-60: report and review

51-54: task 7, report, plot captions and three diagrams. 55-56: task 8, the 30-45-minute meeting, comments and clarified actions. 57-58: reproduce one review issue from data, fix a significant problem or document a limitation. 59-60: repeat affected tests and update rc2. If the reviewer is unavailable, use these hours for independent preparation and rehearsal; move the actual external review to available time in the overall plan with explicit hour accounting.

Distinguish correction from response: saying addressed does not close an issue if the matrix lacks a new result. A comment may change a requirement, but retain the previous wording, reason and new status. This matters particularly for hand loading, admissible slope and claimed autonomous runtime.

Hours 61-75: presenting professional results

61-62: choose a work direction and map competencies to completed work. 63-64: select strong CV evidence, interview questions and the next technical gap. 65: check exit criteria and the G4 list. These 5 analysis hours are split into 2 + 2 + 1.

66-68: task 9 on page 11, portfolio index and CV adaptation. 69-70: task 10, professional feedback or a prepared open-source change with a test. 71-72: final rc2 reproduction and link/package checks. 73-74: close objections, agree limitations and plan G4. 75: decide RELEASE READY/HOLD and prepare the publication package. Public publication is a separate deliberate action after contents review; automatic deployment from a study script is unnecessary.

G3 after M24 is a separate final audit of perception and SLAM, planning recovery, manipulation, ML data and learned-model deployment. Its late date does not retroactively authorise learned-policy hardware control or a Jetson purchase. Retain the classical controller until technical checks pass, and explicitly identify learning tested only in simulation.

Acceptance, professional results and the separate G4

Task 9. A portfolio for the chosen role

Choose one direction: robotics software, embedded-systems and control integration, or machine perception. Match typical work in that direction to your own contributions: driver, estimator, controller, experiment, CAD integration or data processing. For each strong CV point, state a concrete action, conditions, measured result and link to the work. If a result is not yet available, do not invent an improvement percentage; describe the implemented and verified function.

Save `portfolio_index` and a CV version with 3-5 substantiated technical points. Check that every point has a file, test or documented decision behind it and makes your own contribution clear. Boundary case: a ready-made ROS package was used, but the wording credits you with its whole algorithm. Separate integration, your modifications and the third-party component. Prepare 10 interview questions from actual project difficulties and answer them using data.

Task 10. Feedback and the next focus

Choose one small useful change to an open-source project you use, or obtain additional professional feedback on your result. For a change, prepare issue reproduction, the change and a suitable test or documentation; submit separately after owner review. For feedback, save technical comments and actions. Do not make completion of the month depend on a maintainer accepting the contribution; state its actual status.

Save `contribution_or_feedback.md` and an 8-12-week specialisation plan: one gap, one project, a result criterion and time limits. Check that the direction follows an identified weakness or the chosen role. Boundary case: ten new technologies are listed without a practical question. Reduce this to one next verifiable result, such as improved contact observability or hand-force calibration.

M24 exit criteria

Complete the requirements matrix, valid metrics, raw-data package, failure regression checks, setup and maintenance instructions, versioned candidate, technical report and comments actually received from an external reviewer. Label the release READY only when mandatory requirements for the claimed operating scope are closed. Otherwise retain HOLD with a concrete action list. The release draft and index can be fully prepared before public publication.

Pass to the separate G4 the completion of test tables and statistical analysis, independent E-stop and final-state verification, external review with corrections, release archive and future plan. Its 35 hours comprise 7 for diagnostics, 12 for integration, 9 for failure checks and corrections, and 7 for review and decision. PASS permits continuation; CONDITIONAL requires a named correction and responsible person; FAIL means repeating failed work and revising the calendar. Do not silently transfer the hand's OPT work here. The final target for both systems is 22.12.2028; a date does not replace evidence.

Assigned reading: engineering release

The references may be in English; the exercises and definitions are presented here in English. Read only the named sections before the associated work, within the existing 75 hours.

Core reading: three guides

- 1. NASA - Systems Engineering Handbook.** Appendices C, D, E: requirement wording, verification and validation matrices. Then 5.3-5.5: Verification, Validation, Transition. For tasks 1 and 3-6, link requirements, procedures, results and the handoff package for both devices.
- 2. S. Chacon, B. Straub - Pro Git, 2nd ed.** 2.6 Tagging; 7.12 Bundling. For task 5, freeze the exact version and verify repository restoration. Reference: 7.4 Signing Your Work; a signature and verifiable identity differ from a matching hash alone.
- 3. NIST/SEMATECH - e-Handbook of Statistical Methods.** 7.2.4.1 Confidence intervals. For the test campaign, study a success-proportion interval, small samples and the edge cases 0/N, N/N. Record assumptions and check the calculation against the log of all attempts.

Documentation for the selected tool

pytest. How to use fixtures: teardown/cleanup. For package checks, eliminate dependence on test order and leftover files. Automated reproduction tests must not energise physical motors.

Sphinx. Getting started: sources, document structure. For the report, connect the index, instructions and API. Use only if it is already the chosen generator; changing a working stack at programme end is unnecessary.

DVC. Get Started: Tracking data, Retrieving data, Switching between versions. Reference for task 5 if using this data-storage method: restore the correct-version raw data into an empty directory. A .dvc file without accessible data does not complete the archive.

Reading deliverable

Create one matrix: claim → requirement → procedure → raw data → calculation → decision, and reproduce a selected result using the instructions without the old environment. Tests of your assembly establish physical limits; the v1.0 label does not.

Terms and definitions

Requirement. A verifiable mandatory system property with unambiguous conditions and a criterion; a wish without an evaluation method is not a complete requirement.

Acceptance criterion. A predefined PASS/HOLD/FAIL rule for a requirement, including limit, conditions, procedure and required evidence volume.

Verification. Obtaining evidence that a stated requirement is met; methods may include analysis, inspection, demonstration or testing of a specific configuration.

Validation. Obtaining evidence that a system suits the declared application under expected conditions; distinct from compliance with an internal specification.

Traceability. Linking a claim to a requirement, implementation, procedure and result, allowing the effects of changes and the source of conclusions to be checked.

Evidence package. A consistent set of requirements, configurations, procedures, raw data and calculations sufficient to verify specific project claims.

Calculation reproduction. Obtaining a result again from specified data, code and parameters in a documented environment; allowable numerical differences are declared in advance.

Raw data. Primary saved measurements before subsequent processing; they must include units, time, source and recording conditions.

Data provenance. A record of data sources, acquisition conditions and transformations, linking a final metric to the original measurement and processing versions.

Release manifest. A machine-readable list of files with purposes, versions or checksums, used to verify that release contents are complete.

Content hash. A deterministically computed data fingerprint for detecting change against a trusted value; it does not by itself establish authorship or measurement validity.

Digital signature. A cryptographic link between a message and a private key; attribution to an author also requires a trusted association between the public key and identity.

Regression check. Rechecking previously verified behaviour after a change to detect unintended deterioration in the covered scenarios.

Fault injection. Deliberately reproducing a specified fault in a controlled test to check detection, response and the system's final state.

Confidence level. The proportion of intervals covering the true parameter when the procedure is repeated under its assumptions; not the probability that the parameter lies inside an already computed interval.

Selection bias. A systematic difference between attempts included in analysis and the target population, such as excluding failures; increasing N does not automatically remove it.

Relationship: estimated success proportion $\hat{p} = k/N$; also report N, interval method and trial conditions. READY status does not replace this information.

Preparing for release and G4

USE EXISTING EQUIPMENT

Use the MacBook Pro M4, working Ubuntu 24.04 environment with ROS 2 Jazzy, onboard computer, M23 six-legged robot, tools and instruments. For the hand, use the assembly actually built and data from the separate OPT 120 workstream; record their status independently of leg success.

PURCHASES NOW

No new mandatory equipment. Consumables and replacements for defective parts may follow the fault log and existing BOM; additional actuators, a new camera or GPU are unnecessary for release. Use existing storage with verified restoration for the archive.

INSTALL NOW

No new mandatory stack: restore pinned Git, Python, pytest, SciPy and the selected documentation generator. Prepare the report on macOS ARM; run ROS tests in the existing Ubuntu ARM64 environment or on the agreed onboard x86 platform. Record the architecture for every reproducible result.

If the report already uses [Jupyter Book](#), install the pinned version and follow Execute & Compute and Export in the matching documentation without mixing generations. DVC and Sphinx remain conditional components of the chosen workflow; the existing storage and build methods may be retained.

INSTALLATION AND RESTORATION CHECK

In a clean directory, restore the Git-tagged version and data, install dependencies from the lock file, and run regression checks and replay with hardware-command output disabled. Recreate a key table and plot without saved cell outputs. Check [SciPy: proportion_ci](#), manifest contents, raw-data availability and checksums against the trusted manifest.

Open the final PDF, CAD and instructions in another viewer; check links and units. For substantive review, use [Google: Design, Functionality, Complexity, Tests](#). Prepare a [GitHub Release draft](#) with attachments and limitations; publication is a separate owner decision.

HOURLY BOUNDARIES AND FINAL CHECKPOINTS

M24 ends on 27.11.2028 and contains 75 hours. It is followed separately by **G3, 27.11-10.12.2028 - 35 hours**, and **G4, 10-22.12.2028 - 35 hours**. G4 receives verification of both systems, independent stopping, comments and the archive. The hand's OPT has another separate 120 hours; software installation and presentation do not close failed physical acceptance.

Preparation and checks fit within the existing 75 hours, replacing repeated reading and setup. Delivery waiting time does not count as study time.

Motion fundamentals: justifying the robot model

Required synthesis: explain geometry, forces, motion and measurement

By programme end, you should independently reconstruct the calculation chain: foot or finger target → admissible joint coordinates → trajectory over time → required torques → actual actuation → measured motion. Include one such analysis using leg data and one using finger data in the report; reuse completed experiments without creating a new test programme.

A reference example without a robotics library

A planar mechanism's revolute joint is at the origin, with x rightwards and y upwards. A point mass $m=0.40$ kg is attached to the end of a massless link $L=0.30$ m. Angle q is measured from the downward vertical towards +x; neglect friction. For $q=\pi/6$, $\dot{q}=0.40$ rad/s and $\ddot{q}=0.80$ rad/s²: $p=(L \sin q, -L \cos q)=(0.15000; -0.25981)$ m.

Differentiate the geometry: $\dot{p}=(L \cos q, L \sin q) \cdot \dot{q}=(0.10392; 0.06000)$ m/s. Moment of inertia about the joint is $I=mL^2=0.036$ kg·m². Dynamics $I \cdot \ddot{q} + m \cdot g \cdot L \cdot \sin q = \tau_{\text{act}}$ with $g=9.81$ give $0.0288 + 0.5886 = 0.6174$ N·m. The first term accelerates the mass; the second compensates gravity. A link with distributed mass has different I and gravity terms.

Add a vertical environmental force $F_{\text{ext}}=(0;2)$ N at the end. Its torque about the joint is $\tau_{\text{ext}}=p_x \cdot F_y=0.30$ N·m. The equation $I \cdot \ddot{q} + m \cdot g \cdot L \cdot \sin q = \tau_{\text{act}} + \tau_{\text{ext}}$ now requires $\tau_{\text{act}}=0.3174$ N·m. The environment assists motion; do not choose the sign of $J^T F_{\text{ext}}$ from memory without defining force direction.

Four defence questions

Geometry: where do FK/IK and the solution branch come from, and what does the Jacobian lose at a singularity? **Dynamics:** which forces are included, and what changes with mass or speed?

Control: which signal does the actuator actually accept, and why might the requested trajectory be infeasible? **Contact:** what confirms support and grasp, and when does a static model cease to describe the experiment?

Prepare `movement_defense.md`: an axis diagram, notation and units, derivation, numerical hand-calculation results, an independent software calculation, one real log and model limits. Recheck velocity using a central difference of FK, the external-force sign using $r \times F$ and substitution of torque into forward dynamics. This examines connections between modules rather than memorised formulas.

Within 75 hours: 1 hour from 43-44 for the model diagram, 1 hour from 49-50 for rehearsal, and 1 hour from report work in 51-54 for calculation and evidence (task 7). G3, G4 and OPT retain separate hours. Review Lynch, Park, [Modern Robotics §5.2](#) and [Tedrake: Definitions](#); derive this example's equations independently from the sum of torques.